

Learning to Simulate Crowds with Crowds

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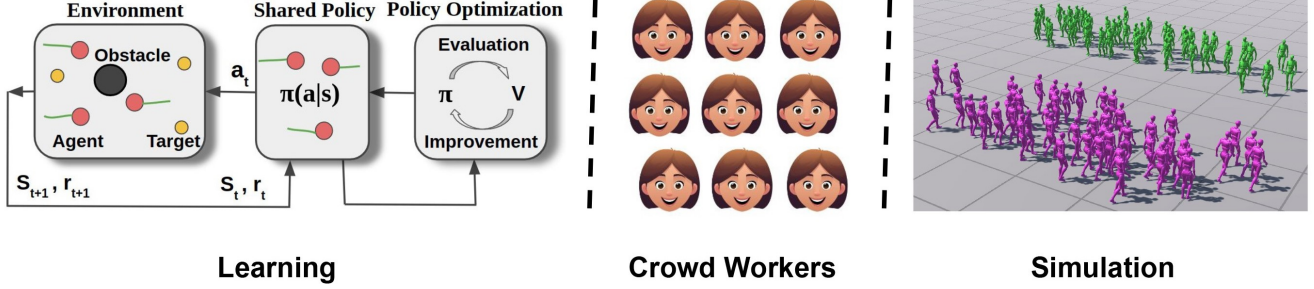


Figure 1: We propose a novel approach for simulating crowd with Deep Reinforcement Learning. Learning optimal crowd navigation behaviors is driven by crowd-sourced feedback, which drives optimal parameters and reward for an RL policy. Finally, we simulate the agents with the policy trained with the optimal learning parameters suggested by the crowd-sourced Bayesian approach.

ABSTRACT

Controlling agent behaviors with Reinforcement Learning is of continuing interest in multiple areas. One major focus is to simulate multi-agent crowds that avoid collisions while locomoting to their goals. Although avoiding collisions is important, it is also necessary to capture realistic anticipatory navigation behaviors. We introduce a novel methodology that includes: 1) an RL method for learning an optimal navigational policy, 2) position-based constraints for correcting policy navigational decisions, and 3) a crowd-sourcing framework for selecting policy control parameters. Based on optimally selected parameters, we train a multi-agent navigation policy, which we demonstrate on crowd benchmarks. We compare our method to existing works, and demonstrate that our approach achieves superior multi-agent behaviors.

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1 INTRODUCTION

Simulating the motion of crowd has implications in domains ranging from robotics to computer graphics applications. Despite much recent progress [Toll and Pettré 2021], simulating such motion is challenging since each crowd agent takes independent navigational decisions that affect the present and future dynamics of other agents,

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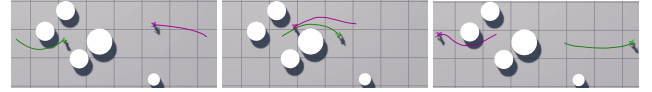


Figure 2: Left to right: Agents locomoting in an environment containing multiple obstacles.

and the environment. To answer such challenges, researchers have taken interest in Learning-based approaches that learn navigational behavior directly from data and continuous experimentation. One such framework is Deep Reinforcement Learning (DRL).

DRL allows, in theory, learning what are the optimal navigation actions each agent should take. Such set of decisions which is known as a navigation policy, is learned through a training regimen that rewards agents for optimal navigational actions. Existing researches in multi-agent navigation with RL have progressed our understanding of this problem domain [Kwiatkowski et al. 2022]. Regrettably, typical approaches focus on imitation of crowd dynamics [Lee et al. 2018; Panayiotou et al. 2022]. These might be sufficient for some applications but lack real-world crowd's features, such as preserving personal space, and locomotion smoothness.

We address these deficits with a crowd-based RL for learning realistic multi-agent dynamics. Key contributions of our work are a novel reward function for realistic agent navigation, embedding position-based multi-agent dynamics as a part of the Bayesian optimization learning process, and a framework for learning optimal reward parameters.

2 METHOD

In our approach, agents in every time step independently select a velocity to take a step forward in time. Our agents use an RL approach to determine an optimal velocity for the next time step. Below, we give more details on the learning process, including our

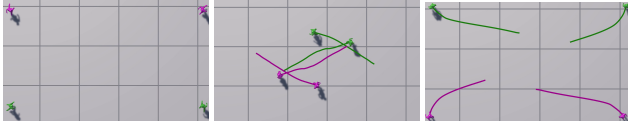


Figure 3: Quad. Left to right: Agents maintain smooth trajectories by performing least-effort maneuvers to avoid collisions.

training and simulation framework, and how we use crowds to select optimal crowd simulation results.

Velocity Selection. In our method, a policy decides on the action a that an agent should take in a state s . Our actions are a velocity-angular velocity tuple (v, ω) where $v \in [-1.3, 1.3]$ and $\omega \in [-\pi/4, \pi/4]$. On the other hand, state s describes the configuration of the environment, which includes agents and obstacles. Agents record the relative positions of neighboring agents and obstacles using visual sensors, along with their relative positions to their individual locomotion goal. Sensors use depth maps and rays.

Rewards. For learning navigation behaviors, our method relies on scalar rewards $r \in \mathbb{R}$ that indicates the quality of a navigational action a taken by agents. Our rewards consist: i) *Distance to goal*, which incentivizes agents to navigate towards their locomotion goal, ii) *Collision avoidance*, to penalize overlapping agents, iii) *steering quality*, which penalizes large changes in velocity, angular velocity, and acceleration, iv) *Personal Space maintenance*, which adds a safe distance from other agents in addition to avoiding collisions.

Simulation Environment. We use Position-Based Dynamics (PBD) as a simulation environment to correct unrealistic navigation decisions. For example, if an agent mistakenly navigates into a wall, in addition to penalizing this decision, we correct it by moving the agent to a correct position. We use the short-range collision avoidance constraint [Weiss et al. 2017] for resolving such situations. PBD adjusts agents' predicted positions $\mathbf{x}$ whenever $\|\mathbf{x}_{ij}^*\| \leq (r_i + r_j)$ and a collision occurs. Here, i and j are agents, and $\mathbf{x}_{ij} = \mathbf{x}_i - \mathbf{x}_j$.

Learning from Preferences. Tuning control parameters for multi-agent simulation is difficult since perception of realistic navigation varies from user to user. Hence we conduct a user study to find the optimal policy. We use Bayesian optimization to estimate the best policy from user selections [Koyama et al. 2017]. In brief, we show users pairs of videos depicting a crowd navigation, where each video results from a different policy, and ask users to select more realistic pair member. Our policy space is generated from augmenting reward values with respect to acceleration, collision, and personal space. A total of 148 participants took part in our survey, from which we collected 1524 responses. In terms of implementation, we use the optimization library provided by [Koyama et al. 2017]. On the preferred policy, we train a navigation policy based on the optimal control parameters inferred from user choices.

Implementation. For the training, we used a fixed number of 10 agents and 6 obstacles while randomizing their starting and goal locations. We experimentally found these numbers optimal for superior agent learning. Agents are trained with control parameters inferred from user choices. We use Proximal Policy Optimization (PPO) with Stochastic Gradient Decent (SGD).

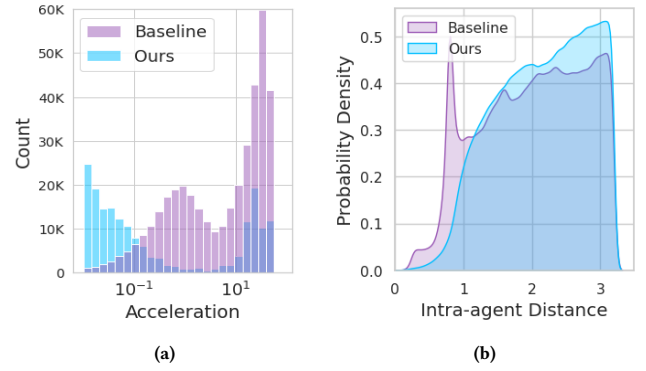


Figure 4: Crowd navigation metrics. Compared to an existing RL work [Lee et al. 2018], agents using our policy show lesser acceleration and greater inter-agent distance, which signals more realistic navigation behaviors.

3 RESULTS AND DISCUSSION

We proposed a learning-based approach for crowd simulation where optimal parameters are inferred via a crowd-sourcing. We demonstrated our method on common crowd simulation scenarios while varying the number of agents from 2 to 100 (Figure 2), and compared our agent navigation behaviors with both existing non-RL algorithms, such as ORCA and Social force [Toll and Pettré 2021], and a RL baseline work [Lee et al. 2018]. Overall, agents using our policy generated smoother trajectories, maintained more intra-agent distance (Figure 4b), arrived at their destination sooner, and had fewer steering changes which result in unwanted acceleration (Figure 4a).

There are several avenues for future work. First, we are interested in simulating diverse pedestrian scenarios that showcase heterogeneous agent behaviors, in terms of realism either in navigation behavior or in physical properties. Second, our framework is integrated with PBD which naively allows various constraints for mimicking complex behaviors such as grouping with minimal runtime cost. Finally, using a dynamic environment during training like integrating an adversarial learning framework for generating previously unseen scenarios could also help agent learning further.

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